Weekly Progress Report

Amity School of Engineering & Technology

Project Group ID -- **Project Area –Drone Tech.**

B.Tech (CSE DS) IV Semester

WPR No:2 **Duration:** 20/05/2025**to** 26/05/2025

Students' Weekly Progress Report (WPR) for NTCC Semester of session 2025-26

To be filled by Students	
Students Name	Pavika Yadav
Roll no.	076
Enrollment no.	A023167023076
Project Title finalized, if Yes, give name, if NO, give reason	Yes, Object detection using Drone technology
Synopsis submitted	yes
Literature review	Prepared the draft
Technical & Economical Feasibility	Not yet done
Bill of Material	Draft prepared
Project Progress Schedule (PERT Chart)	Will start upcoming week
Design of critical components	Not started yet
Fabrication work (give %)	0%
Experimental work (give %)	0%
Result and Analysis	Not applicable yet
Report writing	Not started
Signature of students	Pavika Yadav

Work done in this week:

- Initiated **in-depth literature review** focusing on modern object detection methods:

1. Studied key models: **YOLO (v3, v5, v8)**, **SSD**, and **Faster R-CNN**
2. Compared their accuracy, speed, and hardware suitability for drone usage

- Explored **traditional object detection techniques** (Haar, HOG + SVM) for background understanding
- Enrolled in and began an **online object detection course** (focused on SSD and OpenCV)
- Evaluated **Python libraries** for implementation: OpenCV, TensorFlow, PyTorch
- Drafted a comparison chart of object detection models for future reference
- Started feasibility analysis for software implementation and hardware integration

LITERATURE REVIEW (1ST DRAFT)

1. Introduction

Object detection is a fundamental task in computer vision, where the goal is to identify and locate objects within images or video. With the growth of drone technology, combining drones with object detection models has become increasingly useful for applications such as surveillance, crowd monitoring, and search-and-rescue.

2. Traditional Object Detection Methods

Before deep learning became popular, object detection used classical techniques such as:

• Haar Cascades (Viola-Jones Algorithm)
Used for detecting faces and simple objects. It works fast but has poor accuracy and is limited to predefined object types.

• HOG (Histogram of Oriented Gradients) + SVM
Commonly used for detecting pedestrians. It extracts features from images and classifies them using a support vector machine.
Limitations: These traditional methods require a lot of manual feature engineering and fail in complex environments.

3. Deep Learning-Based Methods

Modern object detection uses deep neural networks. The key approaches include:

-YOLO (You Only Look Once)

• Real-time object detection
•Divides images into grids and predicts bounding boxes & classes
• Very fast — suitable for drones
• Versions: YOLOv3, v4, v5, v7, v8 (latest)

Pros: Fast and good for real-time use
Cons: Less accurate for small objects or overlapping objects (in earlier versions)

-SSD (Single Shot Multibox Detector)

• Detects objects in one shot, similar to YOLO
•Works better than YOLO in some cases for small object detection
• Uses multiple feature maps to detect objects at different scales

Pros: Good balance of speed and accuracy
Cons: Not as fast as YOLO

-Faster R-CNN

• Two-stage detection: first generates region proposals, then classifies them
• Very high accuracy
• Slower than YOLO and SSD

Pros: Very accurate
Cons: Too slow for real-time drone use unless optimized

4. Tools & Libraries

explored the following libraries:

• Python: Programming language of choice for AI/ML work
• OpenCV: For image processing & classical detection
• TensorFlow: Framework for training and deploying models
• PyTorch: Easier for prototyping, widely used in research
• Ultralytics YOLOv5/YOLOv8: Pre-trained models easy to deploy and train on custom data

5. Drones in Object Detection: Recent Studies

Several studies and projects have shown that combining drones with object detection can help in:

•Real-time Traffic monitoring (detecting cars, pedestrians)
•Search and rescue missions in disaster-hit areas
•Wildlife monitoring using aerial imagery
•Crowd counting during events

6. Gaps Identified

• Many models are too heavy to run on lightweight processors like Raspberry Pi or Jetson Nano
• Detection in low-light or fast movement is still a challenge
• Training models for custom objects (like garbage, potholes, etc.) requires a lot of data

7. Conclusion

The literature shows that deep learning, particularly YOLO and SSD, are most suited for drone-based object detection due to their balance of speed and accuracy. However, successful deployment depends on careful model selection, hardware compatibility, and environmental testing.

To be filled by Guide (strike off whichever is not applicable)
Performance of students is satisfactory
Performance of students is unsatisfactory
A warning to be issued to student(s) (Name)
Student was not well (Name)
Date Signature of Guide

Date: 26th May 2025